Exponents

Introduction to Exponents and Powers

The exponents tells us that how many times the number should be multiplied. It is called the Exponential form.
is written like this:

Powers with Negative Exponents

The exponents could be negative also and we can convert them in positive by the following method.

$$100^{-3} = \frac{1}{100^3}$$

This shows that for any non-zero negative integers a,

$$a^{-m} = \frac{1}{a^m}$$

where m is the positive integer and a^m is the multiplicative inverse of a^{-m}.

Laws of Exponents

If we have a and b as the base and m and n as the exponents, then

Laws of Exponents	Example
$a^m \times a^n = a^{m+n}$	$7^3 \times 7^4 = 7^{3+4} = 7^7$
$(a^m)^n$	$(7^3)^4 = 7^{3\times4} = 7^{12}$
$\frac{a^m}{a^n} = a^{m-n}, m > n$	$\frac{7^4}{7^3} = 7^{4-3} = 7^1 = 7$
$a^m b^m = (ab)^n$	$7^3 4^3\ (7\times 4)^3 = 28^3$
$a^0 = 1$	$7^0 = 1$
$a^1 = a$	$7^1 = 7$
$\frac{1}{a^n} = a^{-n}$	$\frac{1}{7^3} = 7^{-3}$

Questions

Simplify:

$2^3 \times 2^4$

(a) 2^7 (b) 2^{12} (c) 8×16 (d) 2^1

Simplify:

$(3^2)^3$

(a) 3^5 (b) 3^6 (c) 9^3 (d) 27^2

Find the value of:

$\frac{5^7}{5^3}$

(a) 5^4 (b) 5^{10} (c) 5^1 (d) 5^5

Which of the following is **equal** to 1?

(a) 7^0 (b) 0^7 (c) 1^0 (d) 0^0

Express in exponential form:

$2 \times 2 \times 2 \times 3 \times 3$

(a) $2^2 \times 3^2$ (b) $2^3 \times 3^2$ (c) $2^2 \times 3^3$ (d) $2^3 \times 3^3$

Simplify:

$\left(\frac{2^3}{2^5}\right)^2$

(a) 2^{-4} (b) 2^{-2} (c) 2^{-6} (d) 2^2

Evaluate:

$10^4 \div 10^2$

(a) 100 (b) 1000 (c) 10 (d) 10000

Which number is **greater**?

(a) 3^5 (b) 5^3 (c) 2^8 (d) 4^4

If $a^3 \times a^x = a^7$, then $x =$

(a) 4 (b) 10 (c) 21 (d) 1

Simplify:

$(x^2y^3)^2$

(a) x^4y^6 (b) x^2y^6 (c) x^4y^3 (d) x^3y^5

Evaluate:

3^{-2}

(a) $\frac{1}{6}$ (b) $\frac{1}{9}$ (c) -9 (d) $-\frac{1}{3}$

Simplify:

$\left(\frac{4^3 \times 2^2}{8^2}\right)$

(a) 2 (b) 4 (c) 8 (d) 1

If $a = 2^3$, $b = 3^2$, and $c = 6$, then what is the value of $\frac{ab}{c^2}$?

(a) 2 (b) 3 (c) 4 (d) 1

Express $\frac{1}{10^6}$ in exponential form with a negative exponent:

(a) 10^{-6} (b) 10^6 (c) -10^6 (d) -10^{-6}

Which of the following is **equal** to $(2^3)^4$?

(a) 2^{12} (b) 2^7 (c) 8^4 (d) both (a) and (c)

What is the value of:

$(x^3y^{-2}) \div (x^{-1}y^4)$?

(a) x^4y^{-6} (b) x^2y^2 (c) x^2y^{-6} (d) x^4y^6

Which expression represents a cube of a square?

(a) $(x^2)^3$ (b) $(x^3)^2$ (c) x^6 (d) All of these

If $a = 10^5$ and $b = 10^{-3}$, then $ab =$

(a) 10^2 (b) 10^8 (c) 10^{-2} (d) 10^3